CMSC477: Robotics Perception and Planning

Project 3: The Energy-Aware Logistics Challenge

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Report Due: May 19, 2025

1. Introduction

In this project, your team will develop an autonomous logistics robot designed to clear a "Loading Dock" by sorting and transporting LEGO bricks to unknown delivery zones. Unlike previous projects, this challenge introduces **resource management constraints**. Your robot must successfully navigate a cluttered environment, locate delivery zones, identify targets (LEGO bricks), and manage a limited battery supply to maximize its delivery score.

2. Environment & Perimeter

- **The Perimeter:** The 10' x 10' arena is strictly bounded by **orange traffic cones** and **colored tape**.
- **Boundary Sensing:** Robots must use **YOLO or color segmentation** to identify these boundaries. Colliding with a perimeter cone or tape results in a penalty.
- **Static Obstacles:** An unknown number of **fabric boxes** will be placed at random locations at the start of the session.
- **Delivery Zones (Goals):** There are two goal locations marked by **AprilTags**. One zone is designated for small bricks and the other for large bricks.
- **Unknown Locations:** The coordinates for the goals, the loading dock, and the delivery zones are unknown beforehand. The robot must explore and map the arena to find them.

3. Logistics & Scoring Rules

The robot's primary task is to transport LEGO bricks from the loading dock to the correct AprilTag goal without fully depleting its battery (i.e., battery level = 0%).

Item Type	Point Value	Battery Consumption
Small LEGO Brick	3 Points	20% Reduction per delivery
Large LEGO Brick	7 Points	40% Reduction per delivery

- **Free Actions:** Actions such as mapping, searching, or returning to the dock without a brick do not consume battery.
- **Energy Consumption:** Battery drain occurs only while the robot is actively transporting a brick. The robot starts the session with 60% battery.
- **Depleted Battery Penalty:** Attempting to transport a brick with 0% battery results in the **negative value** of that brick.
  - *Example (Small):* Delivering a small brick on 0% battery yields **-3 points**.
  - *Example (Large):* Delivering a large brick on 0% battery yields **-7 points**.
- **Moving Bricks:** You can expect a human to move:
  - Any brick in the loading dock, even if the robot is approaching that brick.
  - Any brick from the goal zone back to the appropriate loading dock. You will not lose points if a brick is placed back into the loading dock.

4. Recharging Protocol

The robot must return to the **Recharge Station** (marked by a specific AprilTag) before its battery depletes. To ensure a successful charge:

1. The robot must approach the Recharge Tag **head-on**.
2. The head-on approach must begin from at least **one foot away** from the tag.
3. The robot must be within **five centimeters** of the tag and remain **stationary** for **5 secs**.
4. Once completed, the battery is restored to **100%**.

5. Requirements & Grading

The robot must be autonomous! You may reset it if it gets stuck, but no live coding is allowed.

Rubric

Criteria	Points
Mapping Goals and Dock: Map the arena and find unknown goals (5 pts each) and recharge station (5 pts).	15 pts
Mapping Obstacles: Identify obstacles (perimeter cones and fabric box obstacles).	20 pts
Resource Management: Recharge the battery before it hits 0%.	20 pts
Task Efficiency: Deliver 5+ bricks within the 5-minute time limit.	25 pts
Robust Navigation: Touch a cone (-8 pts) or fabric box (-3 pts).	-8 / -3 pts
Delivering Bricks with Dead Battery: Small brick (-3 pts) / Large brick (-7 pts).	-3 / -7 pts
Report	20 pts

BONUS: Dynamic Mapping: Successfully remap and avoid boxes that move.	+15 pts

6. Submission Guidelines

Teams must submit a video demonstration and a PDF report. The report must include:

- **Block Diagram** of the system architecture showing how the system works (from the camera to the decision-making processes).
- **Bird's Eye View Map** of the environment, containing obstacles (cones and fabric boxes), goals (delivery zones), and recharge station.
- **Methodology** explaining the vision pipeline (YOLO/Segmentation), the battery state machine, and mapping strategy. Your pipeline components should reference the different blocks in your block diagram from above.
- **Results** including the total score, time, resets, and battery efficiency.
- **Appendix:** A full code listing.

Collaboration Policy: You may discuss strategies with other teams, but all code and reports must be original to your team. Plagiarism checkers will be used.

7. Competition Guidelines

Each group will be ranked based on the number of points you scored and the time it took for your robot to complete the task. The team with the **highest score** within the **least amount of time** and **resets** wins the competition. Each team has **five (5) minutes** to complete the task, which includes resets.